YouTube AI Agent Engineering Challenges & Lessons Learned

AIMultiple technical assessment | Development retrospective | October 2026

This report is deliberately separate from the repository README. It explains the practical difficulties observed during development, the approaches implemented, and the remaining limitations. Examples are drawn from the project files, saved outputs and the development history; illustrative scenarios are identified as such.

01 Genius scraping: HTTP 403 and unreliable access

What happened

Live scraping runs, including a Michael Jackson album run, encountered HTTP 403 responses. Genius pages can reject automated access, and a successful URL lookup does not guarantee successful extraction.

How the pipeline responded

Lyrics discovery and extraction were separated into genius_finder.py and genius_scraper.py. Apify-based scraping, alternative URL discovery and retry/fallback behavior were used during development. The system also caches successful lyrics to avoid re-requesting them unnecessarily.

Concrete example

> Artist: Michael Jackson | Album: Got to Be There
> Some Genius requests: HTTP 403 -> retry / alternate attempt

Lesson: external websites are not deterministic APIs. A robust scraper needs explicit failures, useful diagnostics, fallback paths and caching. No claim is made that all 403 responses can be bypassed.

02 Song titles and featuring credits did not always match Genius

What happened

Album track lists may contain extra text, such as a guest artist credit, while Genius uses a shorter title or different slug. Exact-string matching can miss an otherwise valid lyrics page.

Concrete example

> Album listing: I Will Still Love You (with Don Philip) Candidate lookup: I Will Still Love You

The Genius finder was extended to generate normalized song-title variants and validate candidate Genius URLs before accepting them. This was particularly relevant to the Britney Spears album workflow.

Lesson: normalize candidate search terms without changing the original track title in the album output.

03 Lyrics content required careful cleaning and encoding repair

What happened

Extracted Genius text may contain section labels, contributor metadata, extra formatting and character-encoding artifacts. A visually small text difference changes token counts and MD5 hashes.

Concrete examples

> Broken apostrophe: donâ€™t -> don’t Genius markup: [Chorus] / [Verse 1] -> removed by cleaner

The lyrics processor normalizes line endings, repairs common mojibake sequences, strips page metadata and section labels, and then tokenizes the cleaned text with tiktoken cl100k_base.

Important remaining distinction: current code calculates the lyrics MD5 on cleaned text. If the assessment expects MD5 of raw scraped lyrics, that behavior does not match the strict requirement. Cleaning is valuable for analysis but raw-source fidelity should be handled separately.

04 Long album runs could stop partway through

What happened

Scraping an entire album involves multiple external requests. A failure on one track should not force the application to start from the first song again.

How it was addressed

The album processor saves per-track progress in a checkpoint and caches successfully retrieved lyrics. On a later run it checks the artist, album and track list before reusing completed work; saved lyrics must be available for a track to be skipped.

Concrete example

> Britney Spears album: 11 tracks Interrupted run -> checkpoint -> resume remaining tracks
> Recorded completed run: 11/11 tracks, 6,249 tokens

Lesson: for costly or failure-prone pipelines, checkpointing is part of the core architecture rather than an optional convenience.

05 Token counts and hash reproducibility

What happened

Word count is not token count. The assessment also required a multi-step embedding-to-MD5 pipeline, where track order and floating-point formatting matter.

Concrete example: Madonna debut album

> Ordered cl100k_base token counts: 618,587,633,458,567,613,549,888 Total = 4,913 | Mean = 614.125

The pipeline joins the ordered counts with commas, obtains a 768-dimensional embedding from nomic-ai/nomic-embed-text-v1.5, formats each value with ten digits after the decimal point, joins the values with commas and hashes the resulting UTF-8 string with MD5.

> Recorded local hash: 2e5a2b9787d9516816da5c74162ad8f4

The lyrics processor additionally checks that decoding its token IDs reconstructs the cleaned lyrics. This helps detect text/tokenization mismatches, but it does not independently verify the scraped source.

Lesson: exact ordering, tokenizer selection, formatting and model version are part of the data contract. The recorded hash should not be treated as a guaranteed result across all environments.

06 Packaging and GitHub submission decisions

What changed

The final repository was kept modular with main.py as the CLI entry point and helper modules for scraping, processing and hashing. Local development test scripts were intentionally removed from the Git-tracked submission; Madonna and Britney JSON examples remained.

Concrete Git history

```
49310c6  Complete YouTube AI Agent with tests and sample results
4a874f8  Remove development tests from submission
```

The test files were removed from Git tracking with git rm --cached, preserving local copies. This simplifies the repository but reduces the ability of reviewers to run the original development test suite from the checkout.

Known compliance trade-off: the original assignment requested a single Python script; the submitted application uses main.py plus supporting modules. Its JSON output may also include internal token diagnostics. These deviations should be assessed transparently, not described as resolved.

07 Practical takeaways

The most significant engineering lessons were: design for external-service failures; normalize search inputs without corrupting output metadata; keep raw and cleaned text conceptually separate; save partial progress; treat tokenizer/model settings as reproducibility inputs; and document submission trade-offs honestly.

Validation scope: recorded local runs covered Madonna (8 tracks, 4,913 tokens) and Britney Spears (11 tracks, 6,249 tokens), alongside development tests. This report does not claim exhaustive cross-platform or live-service validation.